

Azure Virtual Machines Study Guide

1. Azure VM Overview

Azure Virtual Machines provide **Infrastructure as a Service (IaaS)**, allowing users to deploy scalable compute resources in the cloud. They support multiple OS types, architectures, and configurations for workloads ranging from development to enterprise-grade applications.

2. VM Types (Families)

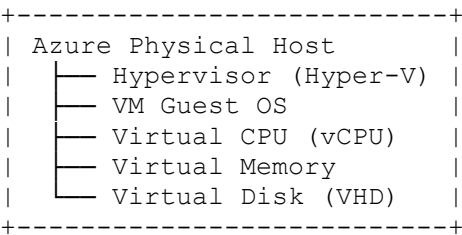
Type	Purpose	Examples
General Purpose	Balanced CPU-to-memory ratio	B-series, D-series
Compute Optimized	High CPU performance	F-series
Memory Optimized	Large memory workloads	E-series, M-series
Storage Optimized	High disk throughput	L-series
GPU	Graphics-intensive workloads	NV, NC, ND-series
High Performance Compute (HPC)	Scientific and engineering workloads	H-series

3. VM Architecture

Azure VMs are built on **Hyper-V virtualization** and run on physical hosts within Azure datacenters.

Diagram: VM Architecture

Code



4. VM Image and Its Types

An **Azure VM image** defines the OS and configuration used to create a VM.

Image Type	Description
Platform Image	Provided by Microsoft (e.g., Windows Server, Ubuntu)
Custom Image	User-created from existing VM configuration
Shared Image Gallery (SIG)	Centralized repository for versioned images
Marketplace Image	Third-party vendor-provided images

5. x86 vs ARM Architecture

Feature	x86 (Intel/AMD)	ARM (Ampere, Graviton)
Instruction Set	Complex (CISC)	Simplified (RISC)
Performance	High single-thread	Efficient multi-thread
Power Efficiency	Moderate	Excellent
Use Case	Legacy apps, Windows workloads	Cloud-native, Linux workloads

6. Redundancy Options

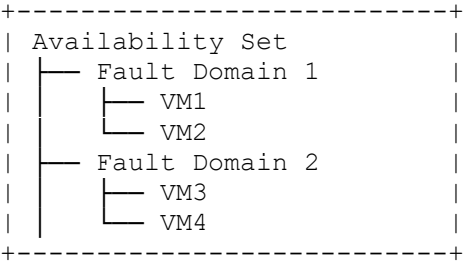
Azure ensures high availability through **Availability Sets** and **VM Scale Sets**.

Availability Sets

- Group of VMs distributed across **Fault Domains** and **Update Domains**.
- Protects against hardware failures and planned maintenance.

Diagram: Availability Set

Code

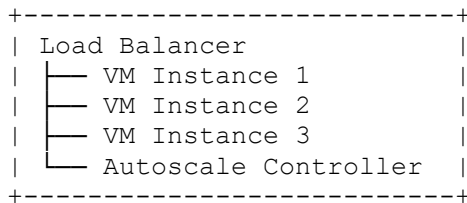


VM Scale Sets

- Automatically deploy and manage identical VMs.
- Supports **auto-scaling** based on CPU, memory, or custom metrics.
- Ideal for stateless workloads.

Diagram: VM Scale Set

Code



7. VM Disk Types

Disk Type	Performance	Use Case
Standard HDD	Low	Dev/test workloads
Standard SSD	Moderate	Balanced cost/performance
Premium SSD	High	Production workloads
Ultra Disk	Extreme	High IOPS, low latency apps
Ephemeral OS Disk	Fast, temporary	Stateless workloads

8. Host Group

- Logical grouping of **dedicated physical hosts** for VMs.
- Used for **isolation, compliance, and workload control**.
- Enables **Host-level affinity** for specific workloads.

9. Capacity Reservations

- Reserve compute capacity in a region for future VM deployments.
- Ensures guaranteed availability for mission-critical workloads.
- Useful for predictable scaling and disaster recovery planning.

10. Performance (NVMe)

- **NVMe disks** provide ultra-fast storage access.
- Used in **Premium SSD v2** and **Ultra Disk** tiers.
- Ideal for databases, analytics, and high IOPS workloads.

11. Proximity Placement Group (PPG)

- Ensures VMs are physically close within the same datacenter.
- Reduces **network latency** between VMs.
- Commonly used for **multi-tier applications** (web, app, DB).

Diagram: Proximity Placement Group

Code

```
+-----+
| Proximity Placement Group |
| ┌── Web VM                |
| │── App VM                |
| └── DB VM                 |
+-----+
```

❑ 12. User Data

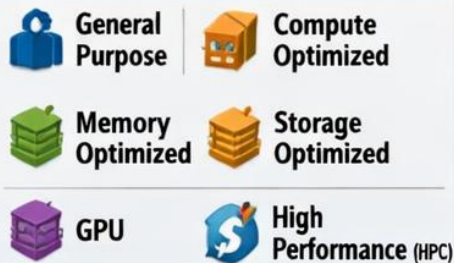
- Custom scripts or configuration data passed during VM creation.
- Used for **initialization**, **software installation**, or **automation**.
- Similar to **cloud-init** in Linux or **Custom Script Extension** in Windows.

✓ Summary Table

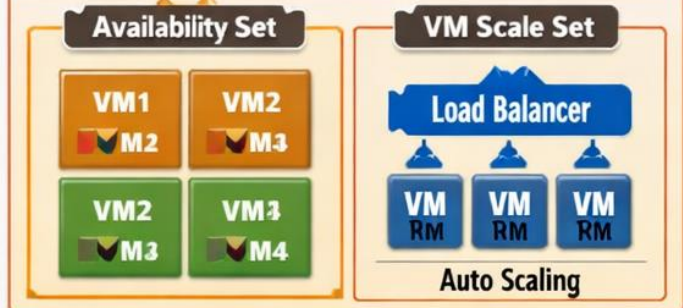
Concept	Purpose
VM Types	Defines workload category
Availability Set	Hardware redundancy
VM Scale Set	Auto-scaling identical VMs
VM Image	OS and configuration template
Host Group	Dedicated physical isolation
Capacity Reservation	Guaranteed compute availability
NVMe	High-speed disk performance
PPG	Low-latency VM grouping
User Data	Custom initialization scripts

Azure Virtual Machines (VMs) Study Sheet

VM Types



Redundancy Options

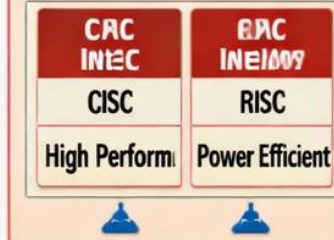


VM Image Types

- Platform Images (Windows, Linux)
- Custom Images
- Shared Image Gallery
- Marketplace Images



x86 vs ARM



ARM - ARM



VM Disk Types

- Standard HDD
- Standard SSD
- Premium SSD
- Ultra Disk
- Ephemeral OS Disk



Host Group & Capacity Reservation



NVMe SSD AEP

High-Speed Storage

User Data

- Startup Scripts & Automation



Proximity Placement Group



Concept	Purpose
VM Types	Workload Categories
Availability Set	Fault Tolerance
VM Scale Set	Auto Scaling
VM Image	OS Templates
Host Group	Dedicated Hosts
Capacity Reservation	Reserved Capacity
NVMe	High Performance Disks
PPG	Low Latency Grouping
User Data	Initialization Scripts